Week-4

Create a native calculator application.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center">

    <EditText
        android:id="@+id/editTextNumber1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter first number"
        android:inputType="numberDecimal" />

    <EditText
        android:id="@+id/editTextNumber2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter second number"
        android:inputType="numberDecimal"
        android:layout_marginTop="8dp" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
```

```
    android:layout_marginTop="16dp"
    android:gravity="center">

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="+" />

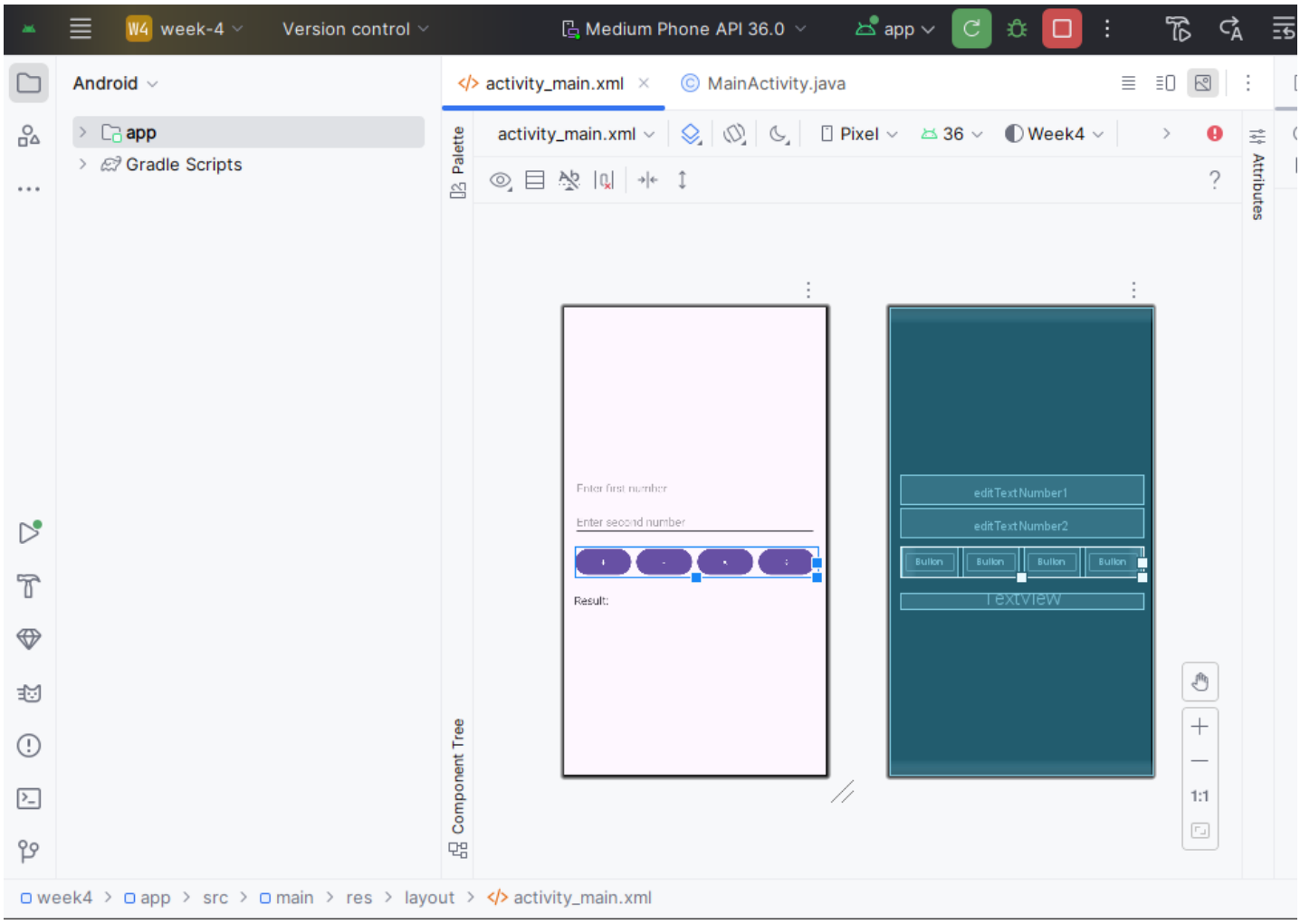
    <Button
        android:id="@+id/btnSubtract"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="-"
        android:layout_marginStart="8dp" />

    <Button
        android:id="@+id/btnMultiply"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="×"
        android:layout_marginStart="8dp" />

    <Button
        android:id="@+id/btnDivide"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="÷"
        android:layout_marginStart="8dp" />
</LinearLayout>
```

```
  <TextView
      android:id="@+id/textViewResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Result: "
      android:textSize="18sp"
      android:layout_marginTop="24dp" />

</LinearLayout>
```



MainActivity.java

```
package com.example.calculatorapp;


import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
```

```
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText editTextNumber1, editTextNumber2;
    Button btnAdd, btnSubtract, btnMultiply, btnDivide;
    TextView textViewResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextNumber1 = findViewById(R.id.editTextNumber1);
        editTextNumber2 = findViewById(R.id.editTextNumber2);
        btnAdd = findViewById(R.id.btnAdd);
        btnSubtract = findViewById(R.id.btnSubtract);
        btnMultiply = findViewById(R.id.btnMultiply);
        btnDivide = findViewById(R.id.btnDivide);
        textViewResult = findViewById(R.id.textViewResult);

        btnAdd.setOnClickListener(v -> calculate('+'));
        btnSubtract.setOnClickListener(v -> calculate('-'));
        btnMultiply.setOnClickListener(v -> calculate('*'));
        btnDivide.setOnClickListener(v -> calculate('/'));
    }

    private void calculate(char operator) {
```

```
String num1Str = editTextNumber1.getText().toString();
String num2Str = editTextNumber2.getText().toString();

if (num1Str.isEmpty() || num2Str.isEmpty()) {
    textViewResult.setText("Please enter both numbers");
    return;
}

double num1 = Double.parseDouble(num1Str);
double num2 = Double.parseDouble(num2Str);
double result = 0;

switch (operator) {
    case '+':
        result = num1 + num2;
        break;
    case '-':
        result = num1 - num2;
        break;
    case '*':
        result = num1 * num2;
        break;
    case '/':
        if (num2 == 0) {
            textViewResult.setText("Cannot divide by zero");
            return;
        }
        result = num1 / num2;
        break;
```

```
        }


        textViewResult.setText("Result: " + result);
    }
}
```

Output:

